Solutions for Exercises on Software Quality Attributes and Scenarios

Description of the Software Product:
You have been hired as a Quality Assurance Specialist to evaluate an ATM application that allows users to perform three banking operations: (i) transfer money between accounts, (ii) deposit cash into an account, and (iii) withdraw cash from an account. The transfer functionality relies on a software service for inter-bank transfers. Below are some requirements for this ATM application:

- **F1:** Users must go through multiple steps of verification before being able to access the main functions of the application. - Security
- **F2:** The ATM's interface should guide users clearly through the process of depositing, withdrawing, and transferring money without needing extensive instructions. - Usability
- **F3**: The application should be capable of managing operations for a large number of users simultaneously without a noticeable impact on the user experience. - Performance (Scalability) / **Note:** Could also be reliability if phrasing the "simultaneously" as a condition
- **F4:** The ATM software must ensure continuous availability and consistent transaction processing during peak banking hours. - Reliability
- **F5:** The software must be adaptable to different types of ATM hardware without requiring significant changes to the codebase. - Portability
- **F6:** The system should facilitate connections with various banking networks for transfer operations and handle potential differences in their protocols seamlessly. - Compatibility
- **F7:** The ATM should process all transactions efficiently to avoid delays, ensuring that users complete their operations promptly. - Performance
- **F8:** The system must have a mechanism to prevent incomplete transactions from causing data inconsistencies and should provide clear feedback when issues occur. - Reliability
- **F9:** Measures should be taken to ensure that all user activities are safely monitored, and inactive sessions automatically end to prevent unauthorized use. - Security
- **F10:** The system's design should allow developers to apply updates or enhancements to specific parts of the code without impacting the entire application. - Maintainability

Part 1 - Quality Attributes and Scenarios

Q1.1 *- You and your development team want to establish some criteria that will enable the system to be more testable. Create a Quality Assurance Plan for the system above.*

Template of the QA Plan:

- *Purpose of the Quality Plan.*
- *Top-four quality attributes. List and justify your choice of the four quality characteristics that should be prioritised in the system. Map those attributes to specific requirements above.*
- *Checklist describing which artefacts and examples of metrics to analyse.*
- *Quality goals per quality attribute (**Tip:** Use the metrics from your checklist).*

Q1.1 Quality Assurance Plan

Please note that the following is an example solution for Exercise Q1.1. You are also encouraged to develop your own alternatives.

Purpose of the Quality Plan

The purpose of this QA Plan is to ensure that the ATM application meets the highest standards of quality and reliability. The plan prioritises specific quality attributes—security, portability, reliability, and performance—to address both user needs and operational demands. The goal is to create a robust ATM application that performs reliably under heavy usage, provides a smooth user experience, and maintains high-security standards.

Top-Four Quality Attributes

1. Security

   Justification: As an ATM system handles sensitive user data and financial transactions, it requires high security to prevent unauthorized access and protect user information.

   Mapped Requirements: F1, F9

2. Portability

   Justification: The application should be able to operate on different ATM hardware without requiring significant code changes. This ensures that the software can function in diverse environments, avoiding the need to develop a new application each time a new ATM model is introduced.

Mapped Requirement: F5

3. Reliability

   Justification: The ATM application must be reliable, particularly during peak hours, to maintain consistent transaction processing and avoid disruptions.

   Mapped Requirements: F4, F8

4. Performance

   Justification: The system needs to support a high number of users simultaneously, process transactions quickly, and scale well under load to avoid user frustration and delays.

   Mapped Requirements: F3, F7

Checklist for Artefacts and Metrics

Attribute	Artefacts	Example Metrics
Security	Access logs, security test results	**Unauthorised access incidents:** Monthly count (goal: 0).
	Session management logs	**Session timeout effectiveness**: Idle sessions end within 2-3 minutes.
	Vulnerability assessment reports	**Risk level:** Identified vulnerabilities rated as “Low” or “None”.
Portability	Compatibility test results, deployment documentation	**Hardware compatibility rate:** Supports 95% of targeted ATM hardware configurations.
	Integration logs and error reports	**Installation time:** Average installation time under 5 minutes for all supported hardware types.
Reliability	Uptime monitoring logs	**System uptime:** Target uptime of 99.9% during peak hours.
	Transaction completion logs	**Transaction success rate:** 99% or higher transaction success during normal and peak hours.
Performance	Load test reports	**Concurrent user support:** Number of simultaneous users supported (goal: 10,000+ without performance drop).
	Transaction response time logs	**Average transaction time:** Transaction processing time < 2 seconds.

Quality Goals per Quality Attribute

Quality Attribute	Goals
Security	Prevent unauthorized access, with monthly unauthorized access incidents at zero.

	Inactive sessions should terminate within 2-3 minutes of inactivity to prevent unauthorized use.
Portability	Achieve a 95% hardware compatibility rate with minimal configuration adjustments required.
	Ensure that the software installs and operates correctly within 5 minutes on different hardware types.
Reliability	Achieve at least 99.9% uptime during peak hours.
	Maintain a 99% or higher transaction success rate with consistent transaction processing.
Performance	Support at least 10,000 concurrent users without noticeable degradation in user experience.
	Complete transaction processing in under 2 seconds on average.

Q1.2 *- Create at least* ***4 quality scenarios*** *using the information from your QA Plan (artefacts, measures, etc.), one for each of the attributes chosen in Q1.1.*

- ***Tip:*** *Use your metrics and goals for the response and response measures.*

Quality Scenarios

1. Security Scenario

Goal: Prevent unauthorized access to the ATM system.

- Source of Stimulus: Unauthorized person attempting to access the ATM.
- Stimulus: A user tries to access the ATM interface after a period of inactivity (e.g., 2-3 minutes) in an unattended session.
- Environment: Production environment, during regular ATM operation hours.
- Artifact: Access logs, session management logs.
- Response: The system should automatically end any inactive session and require re-authentication to resume any actions.
- Response Measure: The system ends the inactive session within 2-3 minutes, logs the session termination event, and prompts for re-authentication, preventing any unauthorized access.

2. Portability Scenario

Goal: Ensure compatibility with different ATM hardware configurations with minimal modifications.

- Source of Stimulus: Deployment of the ATM application to a new type of ATM hardware.
- Stimulus: The application is installed on a newly acquired ATM model that has different hardware specifications from previous models.
- Environment: Testing environment, simulating deployment on a new ATM model.
- Artifact: Compatibility test results, deployment documentation.
- Response: The application installs correctly on the new hardware without requiring significant code changes and passes all functional tests.
- Response Measure: The system achieves a 95% compatibility rate across the new hardware configurations and completes the installation within 5 minutes.

3. Reliability Scenario

Goal: Maintain system reliability and consistent transaction processing, especially during peak hours.

- Source of Stimulus: High user demand during peak hours.
- Stimulus: A peak transaction load occurs, such as when many users attempt to withdraw, deposit, or transfer funds simultaneously.
- Environment: Production environment, specifically during peak banking hours.
- Artifact: Uptime monitoring logs, transaction completion logs.
- Response: The system processes each transaction request accurately and without failure, maintaining availability.
- Response Measure: The ATM system achieves 99.9% uptime and a transaction success rate of 99% or higher during peak hours.

4. Performance Scenario

Goal: Support a high number of concurrent users and ensure efficient transaction processing.

- Source of Stimulus: High transaction volume from multiple users accessing the ATM system concurrently.
- Stimulus: 10,000 users attempt to access the ATM system at the same time for various transactions (withdrawal, deposit, transfer).
- Environment: Production environment during periods of high concurrent use.
- Artifact: Load test reports, transaction response time logs.
- Response: The system handles all transactions without any performance degradation and maintains quick response times.
- Response Measure: The system supports at least 10,000 concurrent users without performance drops, with each transaction completed in under 2 seconds on average.

Part 2 - Reliability and Performance Efficiency

Your team has collected a lot of measures from the ATM system's execution. There is:

- *A file with customer complaints from failures with the respective dates.*
- *A table with measures for each execution of three different features: Transfer, Deposit and Withdraw money.*
- *The timestamps connected to uptime and downtime of the software.*

Tip: *Remember that system failures and crashes are not interchangeable terms when talking about reliability scenarios.*

Q2.1 - *Your team wants to know the POFOD and ROCOF (for the entire month) of the system. For the ROCOF, assume the time period between 2021-01-01 00:00:01 and 2021-02-01 00:00:01.*

POFOD Calculation

Probability of Failure on Demand (POFOD) = failed requests / total requests

1. Total requests is the total number of operations in the file "Performance Log": **11 601**
2. Failed requests is the sum of crashes ("Crashes of the System") and failures ("Customer Complaints"): **35 + 30 = 65**
3. POFOD = 65 / 11 601 = **0.0056**

ROCOF Calculation

Rate of Occurrence of Fault (ROCOF) = failed requests / time

1. Failed requests is the sum of crashes ("Crashes of the System") and failures ("Customer Complaints"): **35 + 30 = 65**
2. The time period between 2021-01-01 00:00:01 and 2021-02-01 00:00:01: **31 days**
3. ROCOF = 65 / 31 = **2.09**

Q2.2 *- Your team wants to know: the Mean Time Between Failures (MTBF), Mean Time to Repair (MTTR) of the system, and the Availability of the system.*

MTBF Calculation

MTBF is the mean duration of the system uptime preceded by downtime.

System downtime and uptime information can be found in the file "Crashes of the System".

System uptime preceded by the system downtime can be calculated by subtracting system up timestamp from the following system down timestamp. For e.g., 2021-01-02 11:21:01 (Row 2: System down) - 2021-01-01 19:02:12 (Row1: System up).
MTFB is the average of the times between failures, which is **62 149 seconds** or **1035.8 minutes**

MTTR Calculation
MTTR is the mean duration of the system being down. System downtime and uptime information can be found in the file “Crashes of the System”.
Repair time = system up timestamp - system down timestamp
MTTR is the average repair time, which is **295.6 seconds** or **4.9 minutes**

Availability Calculation
Availability **=** MTBF / (MTBF + MTTR)
Note: Make sure to have MTBF and MTTR **both** in minutes or in seconds (or any other unit you have selected. The important part is for them to be the same)
Availability = 1035.8 / (1035.8 + 4.9) = **0.995**
Availability is **99.5%**

Q2.3 *- Your team wants to know which of the features are slower. Find the average latency for the deposit, transfer and withdraw operations.*

To find the average latency for the operations, use the file “Performance Log”.
To find out which of the features is the slowest, first, calculate the average latency per feature.
Deposit: 1197.33 ms
Transfer: 400.65 ms
Withdrawal: 7466.16 ms
Withdrawal operation is the slowest because it has the highest average latency.